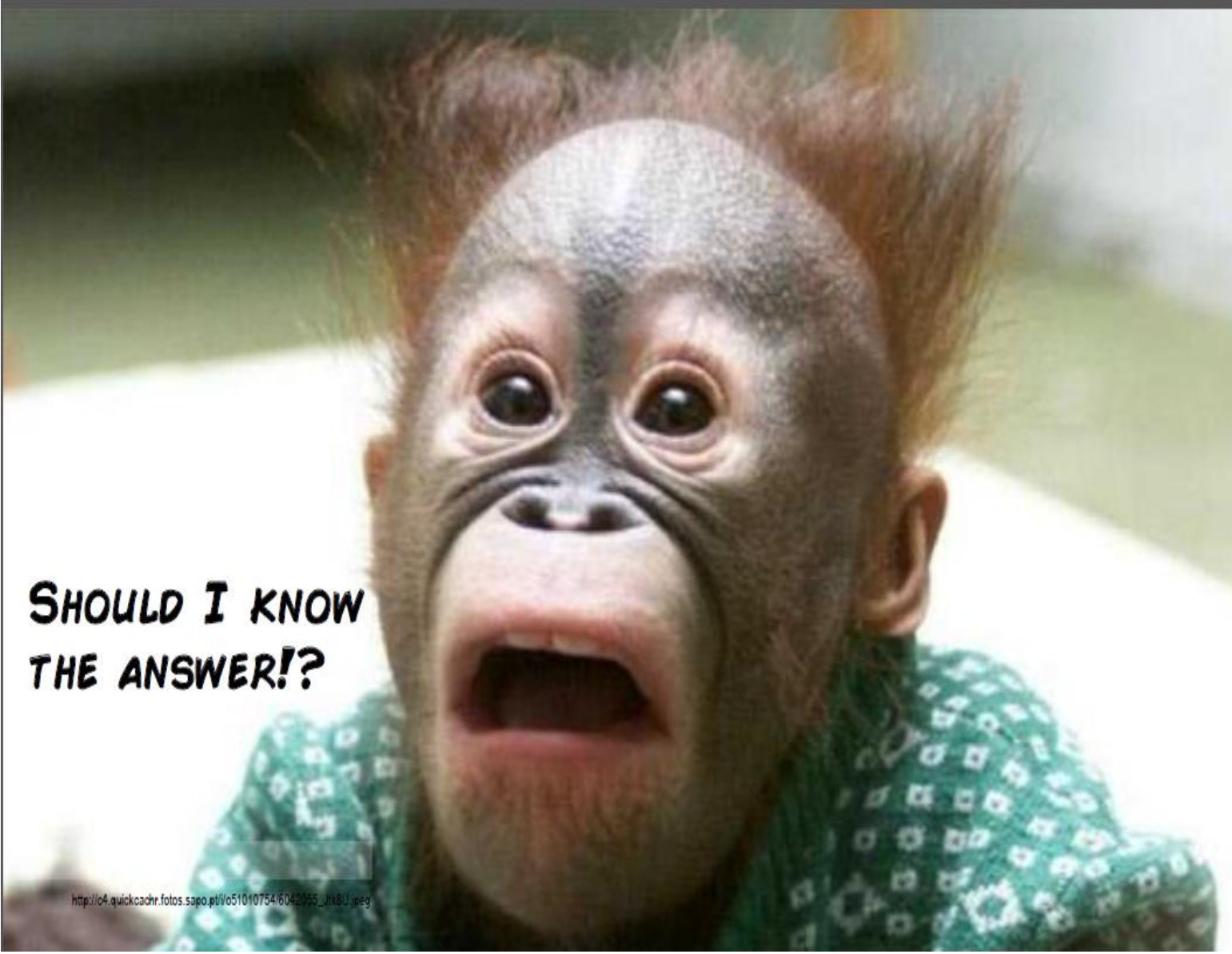


Introduction to Software Engineering

**WHAT ARE YOU
DOING HERE?**

A close-up photograph of a young orangutan. The orangutan has a light brown face with a dark vertical stripe running down the center. Its eyes are wide open, and its mouth is open in a surprised or excited expression. It is wearing a green shirt with a white diamond pattern. The background is blurred, showing some greenery and a white surface.

**SHOULD I KNOW
THE ANSWER!?**

http://e4.quickcache.fotos.sapo.pt/vo51010754/6042055_hx8U.jpg

**IT SAYS HERE
WE WILL LEARN
ABOUT SOFTWARE
ENGINEERING!!**



<http://i0.wp.com/www.nerdglaze.com/wp-content/uploads/2013/08/nerdy-dude.jpg?resize=450%2C305>

Contents.

1. Engineering
2. Software Crisis
3. Software relevance
4. Software Engineering
5. Nature of software
6. Quality and Software Quality
7. Software types and domains



Motivation

Importance of software

Software can have a huge impact in any aspect of society.

Where can you find software?



Some popular ones...



Some popular ones...

facebook

☐ Remember Me

[Forgot your password?](#)

Email

Password

Login

Facebook helps you connect and share with
the people in your life.



Sign Up

It's free and anyone can join

Full Name:

Your Email:

New Password:

I am:

Birthday:

Why do I need to provide this?

Sign Up

To create a page for a celebrity, band or business, [click here.](#)

Some popular ones...

You Tube Broadcast Yourself™
Mundial | Español

(0) **blpgirl2** ▾ | Cuenta | Lista rápida (0) | Ayuda | Salir

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If programmers have make a plane



0:19 / 1:00

★★★★★ 389 puntuaciones Reproducciones: 259253

 **DikiyKaban**
02 de junio de 2007
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And even in...



Conclusion

Software is Almost
Everywhere.

Problems in software development

Common issues

- The final Software doesn't fulfill the needs of the customer.
- Hard to extend and improve: if you want to add a functionality later is mission impossible.
- Bad documentation.
- Bad quality: frequent errors, hard to use, ...
- More time and costs than expected



How the customer explained it



How the Project Leader understood it



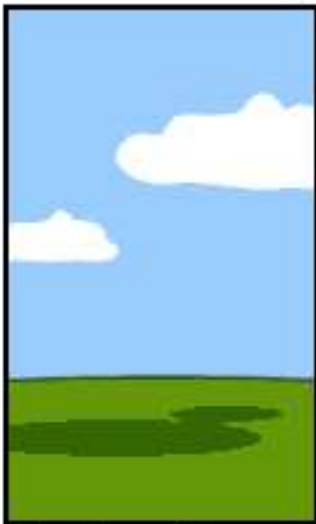
How the Analyst designed it



How the Programmer wrote it



How the Business Consultant described it



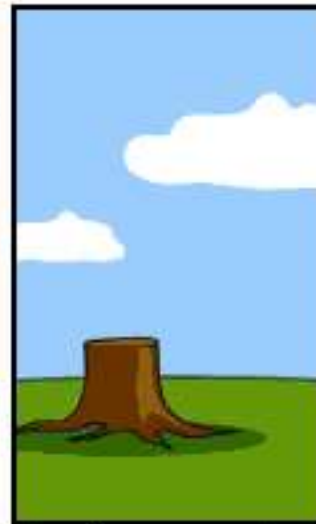
How the project was documented



What operations installed



How the customer was billed



How it was supported



What the customer really needed

But

That never
happens,

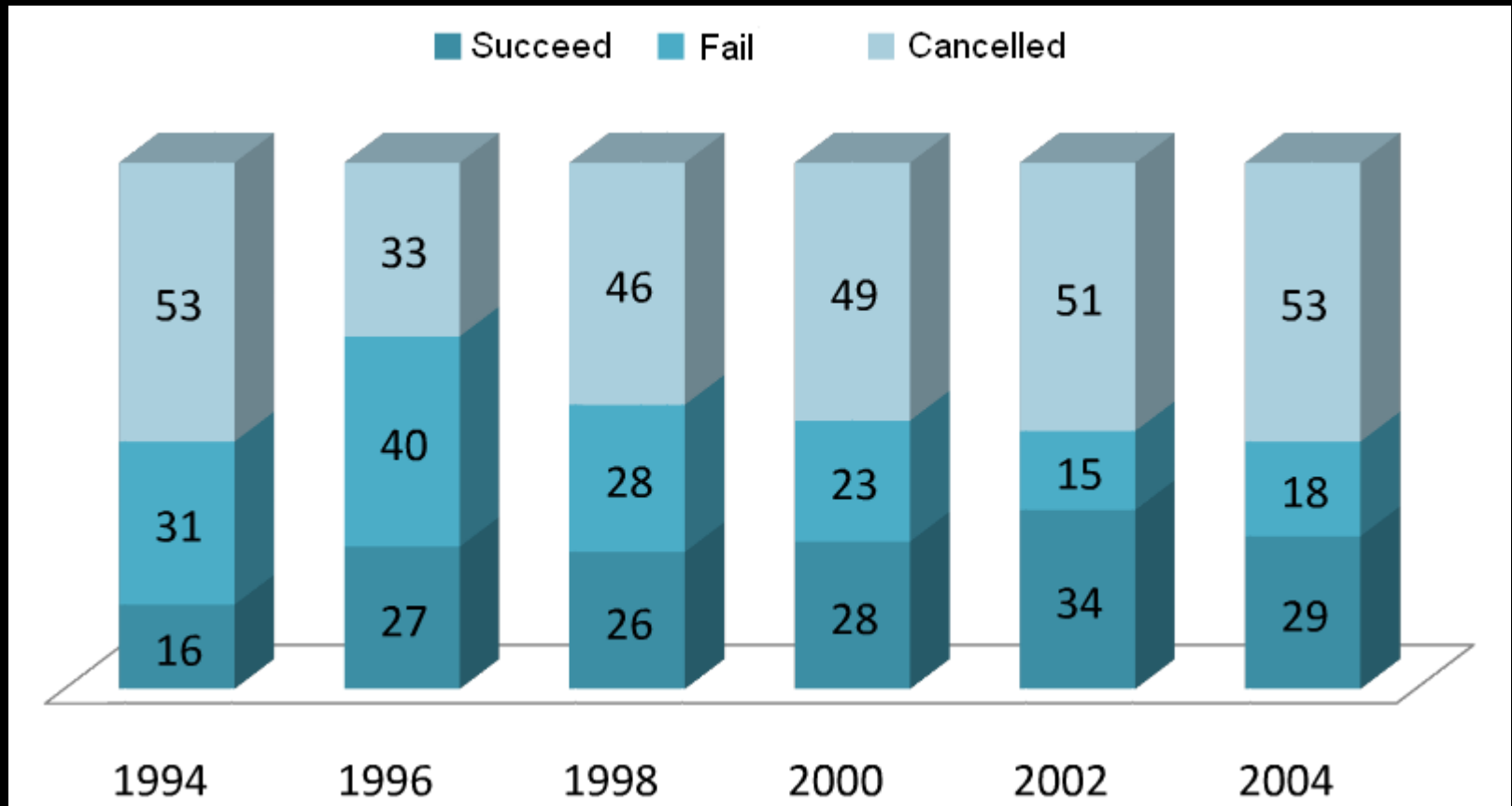
right?

Wrong!

Ariane 5 Flight 501

Cause: design errors in the software

Chaos Report



Conclusion

Programming is NOT enough!

It is not enough to do your best: you must
Know what to do, and THEN do your best.

-- *W. Edwards Deming*

And Since...

A clever person solves a problem.

A wise person avoids it.

- *Albert Einstein*

Solution

Software Engineering

What is it?

The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software, and the study of these approaches; that is, the application of engineering to software.

-Wikipedia

???

Software Engineering

What is it?

The study and application of methodologies to develop quality software that fulfill customer needs.



Software Engineering

Objetive

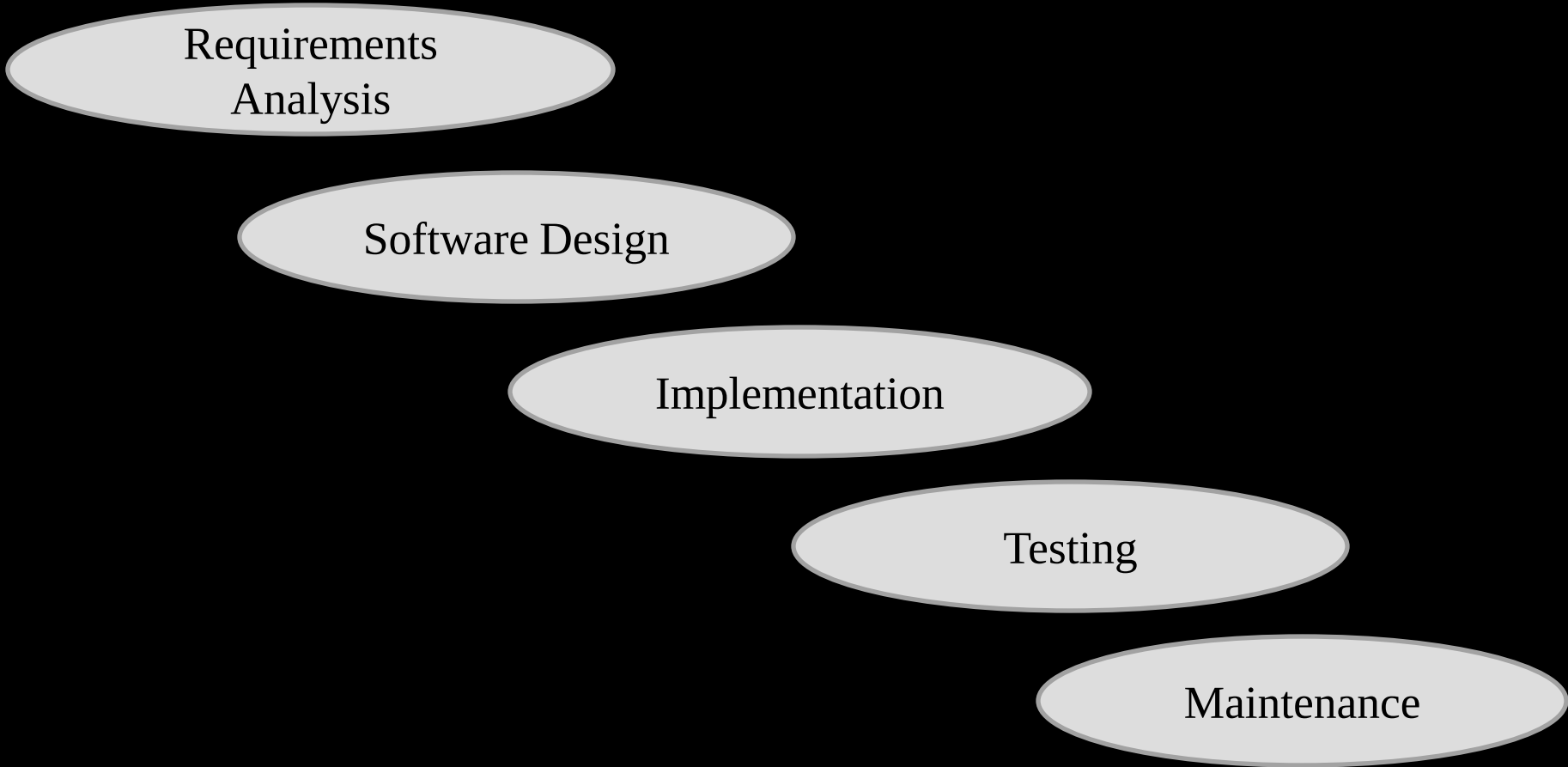
To produce software that is:

- On time: is deliver at the established date.
- Reliable: doesn't crash.
- Complete: good documentation, fulfill customer needs.

The team



Stages for software development



```
graph TD; A[Requirements Analysis] --> B[Software Design]; B --> C[Implementation]; C --> D[Testing]; D --> E[Maintenance];
```

Requirements
Analysis

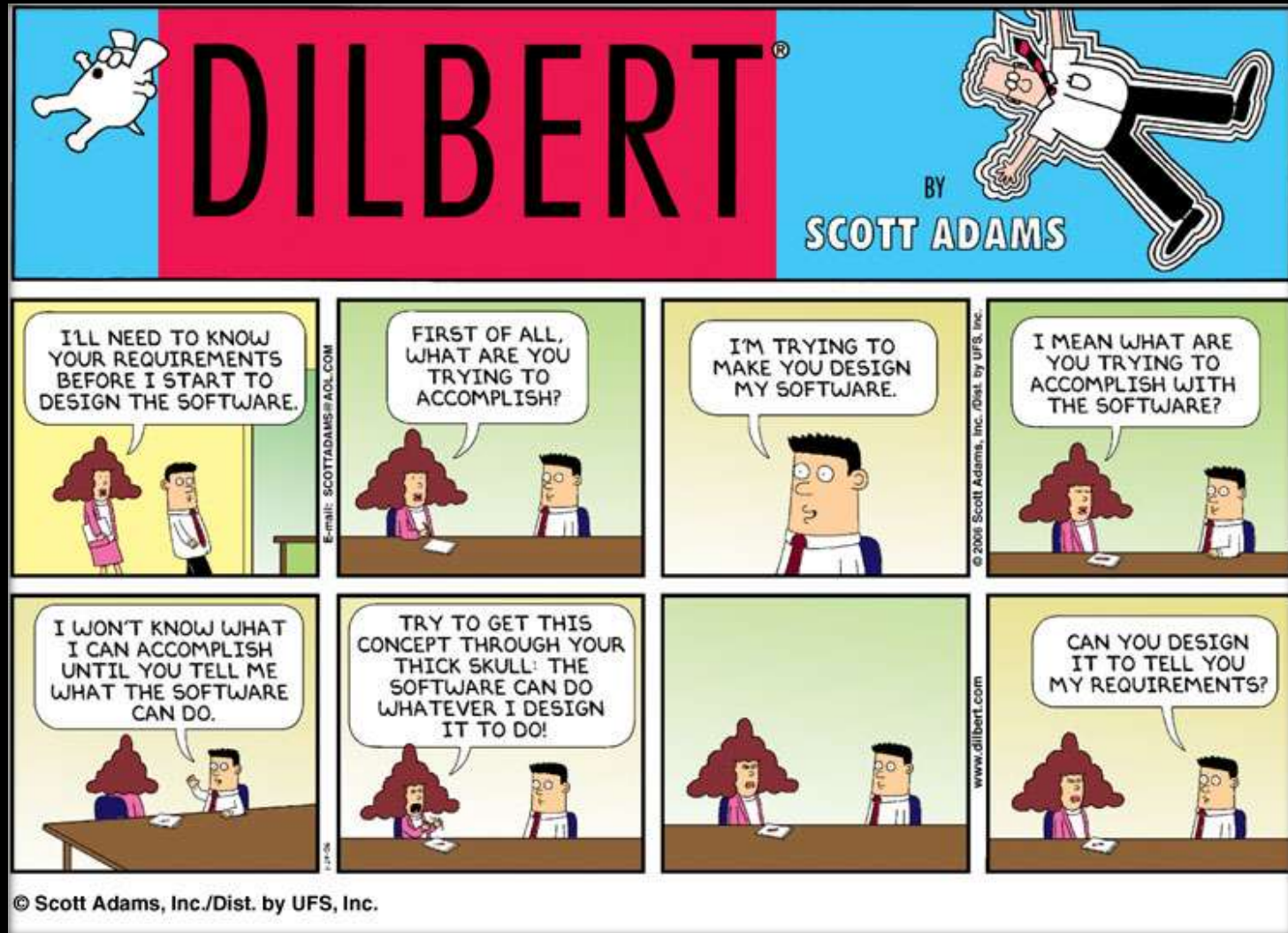
Software Design

Implementation

Testing

Maintenance

1. Requirements Analysis



Find out what the client want the software to do

2. Design

TYPICAL APPLE PRODUCT...

TOUCH

A GOOGLE PRODUCT...

YOUR COMPANY'S APP...

FIRST NAME:		TYPE CD:		4 - K
LAST NAME:		TQP STAT:	☐	AA2-
SSN:		VER:		DK9B
ID:		FT/PT:	☒	KKAP
PHONE 1:		CAT CD:		CN3
PHONE 2:		CITY:		AA-9
ADDR 1:		STATE:		NEW
ACCT #:		ZIP:		DEL
ORD #:				

OKAY

APPLY

SAVE

UNDO

HELP

DELETE

EDIT

SELECT

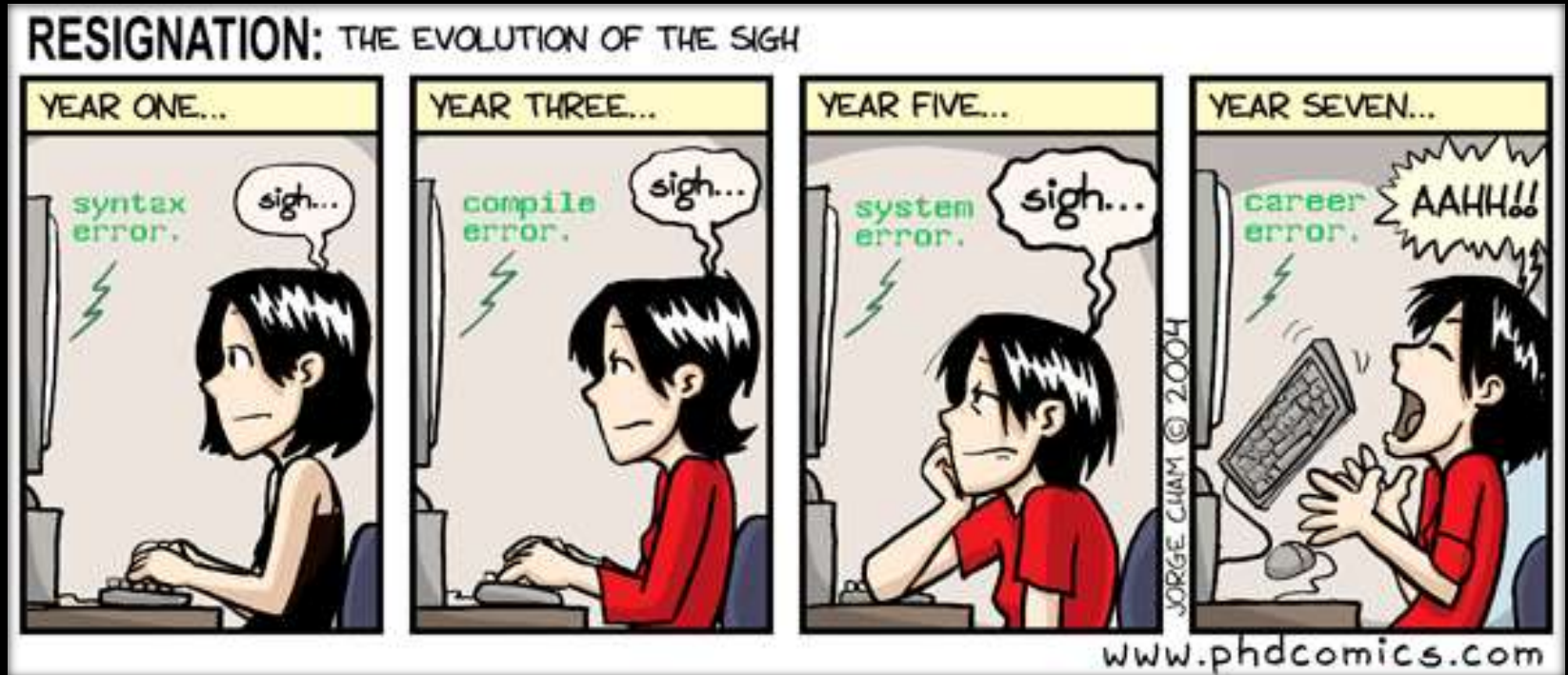
BROWSE

ERRORS

STUFFTHATHAPPENS.COM BY ERIC BURKE

Planning the software solution

3. Implementation



Code!!!

4. Testing



Executing the application trying to find software bugs

5. Maintenance

Any activity oriented to change an existing software product.

What!?

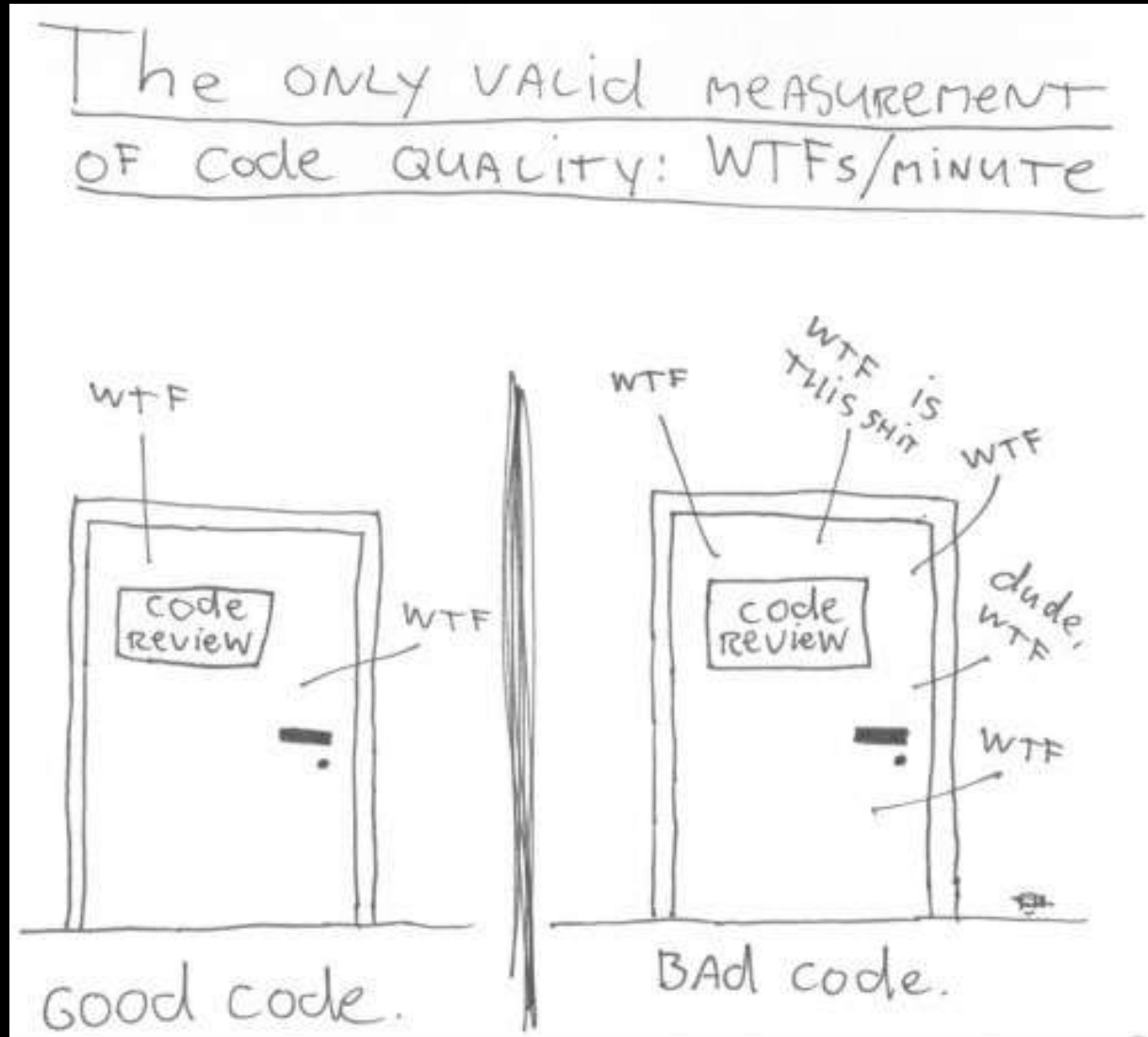
You were expecting another
comic strip?

Well, guess what...

I couldn't find one related ☹️

But

As consolation



Sorry, couldn't resist :P

References

- http://upload.wikimedia.org/wikipedia/commons/thumb/a/a2/Avianca_767-200_at_El_Dorado.JPG/800px-Avianca_767-200_at_El_Dorado.JPG
- <http://www.nosoloviajeros.com/imagenes/colombia/transmilenio.jpg>
- <http://files.nireblog.com/blogs1/keniecita/files/celular-2.jpg>
- <http://www.fayerwayer.com/up/2008/06/iphone3g.jpg>
- <http://www.slideshare.net/soreygarcia/ingenieria-de-software-para-dummies>
- <http://www.projectsmart.co.uk/docs/chaos-report.pdf>
- http://www.evialiens.com/images/software_engineering_explained.gif
- <http://sunnyday.mit.edu/accidents/Ariane5accidentreport.html>
- <http://stackoverflow.com/questions/84556/whats-your-favorite-programmer-cartoon>
- <http://stuffthathappens.com/blog/2008/03/05/simplicity/>
- <http://www.troyangrignon.com/dilbertsoftwarerequirements.jpg>
- <http://www.phdcomics.com/comics/archive/phd120804s.gif>
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